In [10]:
```
# IMPORT LIBRARIES

import pandas as pd
import json
import os
```

In [11]:
```
# LOAD THE DATA AND EXTRACTION FOR AGGREGATED TRANSACTION DATAS

path=r"D:\PHONEPAY_PROJECT\first_project\pulse\data\aggregated\transaction\country\
path = path.replace("\\", "/")
agg_state_trans_list=sorted(os.listdir(path))
print(agg_state_trans_list)

clm1={'state':[], 'year':[],'quarter':[],'transaction_type':[], 'transaction_count'

for i in agg_state_trans_list:
    p_i=path+i+"/"
    agg_yr=os.listdir(p_i)
    for j in agg_yr:
        p_j=p_i+j+"/"
        agg_yr_list=os.listdir(p_j)
        for k in agg_yr_list:
            p_k=p_j+k
            with open(p_k, 'r') as Data:
                D = json.load(Data)
                for z in D['data']['transactionData']:
                    name=z['name']
                    count=z['paymentInstruments'][0]['count']
                    amount=z['paymentInstruments'][0]['amount']
                    clm1['transaction_type'].append(name)
                    clm1['transaction_count'].append(count)
                    clm1['transaction_amount'].append(amount)
                    clm1['state'].append(i)
                    clm1['year'].append(j)
                    clm1['quarter'].append(int(k.replace('.json', '')))

agg_trans=pd.DataFrame(clm1)
print(agg_trans)
```

```
['andaman-&-nicobar-islands', 'andhra-pradesh', 'arunachal-pradesh', 'assam', 'biha
r', 'chandigarh', 'chhattisgarh', 'dadra-&-nagar-haveli-&-daman-&-diu', 'delhi', 'go
a', 'gujarat', 'haryana', 'himachal-pradesh', 'jammu-&-kashmir', 'jharkhand', 'karna
taka', 'kerala', 'ladakh', 'lakshadweep', 'madhya-pradesh', 'maharashtra', 'manipu
r', 'meghalaya', 'mizoram', 'nagaland', 'odisha', 'puducherry', 'punjab', 'rajastha
n', 'sikkim', 'tamil-nadu', 'telangana', 'tripura', 'uttar-pradesh', 'uttarakhand',
'west-bengal']
                         state  year  quarter          transaction_type  \
0     andaman-&-nicobar-islands  2018        1  Recharge & bill payments
1     andaman-&-nicobar-islands  2018        1     Peer-to-peer payments
2     andaman-&-nicobar-islands  2018        1         Merchant payments
3     andaman-&-nicobar-islands  2018        1        Financial Services
4     andaman-&-nicobar-islands  2018        1                    Others
...                         ...   ...      ...                       ...
5029                west-bengal  2024        4         Merchant payments
5030                west-bengal  2024        4     Peer-to-peer payments
5031                west-bengal  2024        4  Recharge & bill payments
5032                west-bengal  2024        4        Financial Services
5033                west-bengal  2024        4                    Others

      transaction_count  transaction_amount
0                  4200        1.845307e+06
1                  1871        1.213866e+07
2                   298        4.525072e+05
3                    33        1.060142e+04
4                   256        1.846899e+05
...                 ...                 ...
5029          655100809        3.892862e+11
5030          493217788        1.361927e+12
5031           76043195        5.753406e+10
5032            2352084        8.472965e+08
5033             421806        5.053641e+08

[5034 rows x 6 columns]
```

In [12]:

```
# LOAD THE DATA AND EXTRACTION FOR AGGREGATED INSURANCE DATAS

path=r"D:\PHONEPAY_PROJECT\first_project\pulse\data\aggregated\insurance\country\in
path = path.replace("\\", "/")
agg_state_ins_list=os.listdir(path)
print(agg_state_ins_list)

clm2 = {'state':[], 'year':[],'quarter':[],'insurance_type':[], 'insurance_count':[

for i in agg_state_ins_list:
  p_i = path + i + "/"
  agg_yr = os.listdir(p_i)
  for j in agg_yr:
    p_j = p_i + j + "/"
    agg_yr_list = os.listdir(p_j)
    for k in agg_yr_list:
      p_k = p_j + k
      with open(p_k, 'r') as Data:
          D = json.load(Data)
          for z in D['data']['transactionData']:
              name = z['name']
```

```
                count=z['paymentInstruments'][0]['count']
                amount=z['paymentInstruments'][0]['amount']
                clm2['insurance_type'].append(name)
                clm2['insurance_count'].append(count)
                clm2['insurance_amount'].append(amount)
                clm2['state'].append(i)
                clm2['year'].append(j)
                clm2['quarter'].append(int(k.strip('.json')))


agg_ins=pd.DataFrame(clm2)
print(agg_ins)
```

```
['andaman-&-nicobar-islands', 'andhra-pradesh', 'arunachal-pradesh', 'assam', 'biha
r', 'chandigarh', 'chhattisgarh', 'dadra-&-nagar-haveli-&-daman-&-diu', 'delhi', 'go
a', 'gujarat', 'haryana', 'himachal-pradesh', 'jammu-&-kashmir', 'jharkhand', 'karna
taka', 'kerala', 'ladakh', 'lakshadweep', 'madhya-pradesh', 'maharashtra', 'manipu
r', 'meghalaya', 'mizoram', 'nagaland', 'odisha', 'puducherry', 'punjab', 'rajastha
n', 'sikkim', 'tamil-nadu', 'telangana', 'tripura', 'uttar-pradesh', 'uttarakhand',
'west-bengal']
                         state  year  quarter insurance_type  insurance_count  \
0    andaman-&-nicobar-islands  2020        2      Insurance                6
1    andaman-&-nicobar-islands  2020        3      Insurance               41
2    andaman-&-nicobar-islands  2020        4      Insurance              124
3    andaman-&-nicobar-islands  2021        1      Insurance              225
4    andaman-&-nicobar-islands  2021        2      Insurance              137
..                         ...   ...      ...            ...              ...
677                west-bengal  2023        4      Insurance            72712
678                west-bengal  2024        1      Insurance            79576
679                west-bengal  2024        2      Insurance            67048
680                west-bengal  2024        3      Insurance            77158
681                west-bengal  2024        4      Insurance            91719

     insurance_amount
0              1360.0
1             15380.0
2            157975.0
3            244266.0
4            181504.0
..                ...
677       100365562.0
678       104987909.0
679        89476633.0
680       107451766.0
681       120602777.0

[682 rows x 6 columns]
```

In [13]:

```
# LOAD THE DATA AND EXTRACTION FOR AGGREGATED USER DATAS

path=r"D:\PHONEPAY_PROJECT\first_project\pulse\data\aggregated\user\country\india\s
path = path.replace("\\", "/")
agg_state_user_list=sorted(os.listdir(path))
agg_state_user_list

clm3 = {
```

```
    'state':[],
    'year':[],
    'quarter':[],
    'brand':[],
    'brand_users':[],
    'percentage':[],
    'registeredUsers':[],
    'appOpens':[]
}

for i in agg_state_user_list:
    p_i = path + i + "/"
    agg_yr = os.listdir(p_i)
    for j in agg_yr:
        p_j = p_i + j + "/"
        agg_yr_list = os.listdir(p_j)
        for k in agg_yr_list:
            p_k = p_j + k
            with open(p_k, 'r') as Data:
                D = json.load(Data)
                registered_users = D["data"]["aggregated"]["registeredUsers"]
                app_opens = D["data"]["aggregated"]["appOpens"]
                if "usersByDevice" in D["data"] and D["data"]["usersByDevice"]:
                    for z in D["data"]["usersByDevice"]:
                        clm3['state'].append(i)
                        clm3['year'].append(int(j))
                        clm3['quarter'].append(int(k.replace('.json','')))
                        clm3['brand'].append(z['brand'])
                        clm3['brand_users'].append(z['count'])
                        clm3['percentage'].append(z['percentage'])
                        clm3['registeredUsers'].append(registered_users)
                        clm3['appOpens'].append(app_opens)

agg_users = pd.DataFrame(clm3)
print(agg_users)
```

```
                          state  year  quarter    brand  brand_users  \
0     andaman-&-nicobar-islands  2018        1   Xiaomi         1665
1     andaman-&-nicobar-islands  2018        1  Samsung         1445
2     andaman-&-nicobar-islands  2018        1     Vivo          982
3     andaman-&-nicobar-islands  2018        1     Oppo          501
4     andaman-&-nicobar-islands  2018        1  OnePlus          332
...                         ...   ...      ...      ...          ...
6727                west-bengal  2022        1   Lenovo       330017
6728                west-bengal  2022        1  Infinix       284678
6729                west-bengal  2022        1     Asus       280347
6730                west-bengal  2022        1    Apple       277752
6731                west-bengal  2022        1   Others      2196334

      percentage  registeredUsers   appOpens
0       0.247033             6740          0
1       0.214392             6740          0
2       0.145697             6740          0
3       0.074332             6740          0
4       0.049258             6740          0
...          ...              ...        ...
6727    0.015056         21919787  236131065
6728    0.012987         21919787  236131065
6729    0.012790         21919787  236131065
6730    0.012671         21919787  236131065
6731    0.100199         21919787  236131065

[6732 rows x 8 columns]
```

In [14]:

```
# LOAD THE DATA AND EXTRACTION FOR MAP TRANSACTION DATAS

path=r"D:\PHONEPAY_PROJECT\first_project\pulse\data\map\transaction\hover\country\i
path = path.replace("\\", "/")
map_state_trans_list=sorted(os.listdir(path))
print(map_state_trans_list)

clm4={'state':[], 'year':[],'quarter':[],'type':[], 'count':[], 'amount':[]}

for i in map_state_trans_list:
    p_i=path+i+"/"
    map_yr=os.listdir(p_i)
    for j in map_yr:
        p_j=p_i+j+"/"
        map_yr_list=os.listdir(p_j)
        for k in map_yr_list:
            p_k=p_j+k
            with open(p_k, 'r') as Data:
                D = json.load(Data)
                for z in D['data']['hoverDataList']:
                    name=z['name']
                    type=z['metric'][0]['type']
                    count=z['metric'][0]['count']
                    amount=z['metric'][0]['amount']
                    clm4['type'].append(z['name'])
                    clm4['count'].append(count)
                    clm4['amount'].append(amount)
                    clm4['state'].append(i)
```

```
                    clm4['year'].append(j)
                    clm4['quarter'].append(int(k.replace('.json', '')))


map_trans=pd.DataFrame(clm4)
print(map_trans)
```

```
['andaman-&-nicobar-islands', 'andhra-pradesh', 'arunachal-pradesh', 'assam', 'biha
r', 'chandigarh', 'chhattisgarh', 'dadra-&-nagar-haveli-&-daman-&-diu', 'delhi', 'go
a', 'gujarat', 'haryana', 'himachal-pradesh', 'jammu-&-kashmir', 'jharkhand', 'karna
taka', 'kerala', 'ladakh', 'lakshadweep', 'madhya-pradesh', 'maharashtra', 'manipu
r', 'meghalaya', 'mizoram', 'nagaland', 'odisha', 'puducherry', 'punjab', 'rajastha
n', 'sikkim', 'tamil-nadu', 'telangana', 'tripura', 'uttar-pradesh', 'uttarakhand',
'west-bengal']
                          state  year  quarter  \
0      andaman-&-nicobar-islands  2018        1
1      andaman-&-nicobar-islands  2018        1
2      andaman-&-nicobar-islands  2018        1
3      andaman-&-nicobar-islands  2018        2
4      andaman-&-nicobar-islands  2018        2
...                          ...   ...      ...
20599                west-bengal  2024        4
20600                west-bengal  2024        4
20601                west-bengal  2024        4
20602                west-bengal  2024        4
20603                west-bengal  2024        4

                                  type     count        amount
0      north and middle andaman district       442  9.316631e+05
1                 south andaman district      5688  1.256025e+07
2                      nicobars district       528  1.139849e+06
3      north and middle andaman district       825  1.317863e+06
4                 south andaman district      9395  2.394824e+07
...                                  ...       ...           ...
20599                alipurduar district  15875637  2.099251e+10
20600         paschim bardhaman district  56616799  6.968735e+10
20601                     nadia district  65274337  1.079320e+11
20602                   birbhum district  36905213  5.778701e+10
20603           purba medinipur district  67833775  1.142771e+11

[20604 rows x 6 columns]
```

In [15]:

```
# LOAD THE DATA AND EXTRACTION FOR MAP INSURANCE DATAS

path=r"D:\PHONEPAY_PROJECT\first_project\pulse\data\map\insurance\hover\country\ind
path = path.replace("\\", "/")
map_state_ins_list=sorted(os.listdir(path))
print(map_state_ins_list)

clm5={'state':[], 'year':[],'quarter':[],'type':[], 'count':[], 'amount':[]}

for i in map_state_ins_list:
    p_i=path+i+"/"
    map_yr=os.listdir(p_i)
    for j in map_yr:
        p_j=p_i+j+"/"
```

```
        map_yr_list=os.listdir(p_j)
        for k in map_yr_list:
            p_k=p_j+k
            with open(p_k, 'r') as Data:
                D = json.load(Data)
                for z in D['data']['hoverDataList']:
                    name=z['name']
                    type=z['metric'][0]['type']
                    count=z['metric'][0]['count']
                    amount=z['metric'][0]['amount']
                    clm5['type'].append(z['name'])
                    clm5['count'].append(count)
                    clm5['amount'].append(amount)
                    clm5['state'].append(i)
                    clm5['year'].append(j)
                    clm5['quarter'].append(int(k.replace('.json', '')))

map_ins=pd.DataFrame(clm5)
print(map_ins)
```

```
['andaman-&-nicobar-islands', 'andhra-pradesh', 'arunachal-pradesh', 'assam', 'biha
r', 'chandigarh', 'chhattisgarh', 'dadra-&-nagar-haveli-&-daman-&-diu', 'delhi', 'go
a', 'gujarat', 'haryana', 'himachal-pradesh', 'jammu-&-kashmir', 'jharkhand', 'karna
taka', 'kerala', 'ladakh', 'lakshadweep', 'madhya-pradesh', 'maharashtra', 'manipu
r', 'meghalaya', 'mizoram', 'nagaland', 'odisha', 'puducherry', 'punjab', 'rajastha
n', 'sikkim', 'tamil-nadu', 'telangana', 'tripura', 'uttar-pradesh', 'uttarakhand',
'west-bengal']
                          state  year  quarter  \
0      andaman-&-nicobar-islands  2020        2
1      andaman-&-nicobar-islands  2020        2
2      andaman-&-nicobar-islands  2020        3
3      andaman-&-nicobar-islands  2020        3
4      andaman-&-nicobar-islands  2020        3
...                          ...   ...      ...
13871                west-bengal  2024        4
13872                west-bengal  2024        4
13873                west-bengal  2024        4
13874                west-bengal  2024        4
13875                west-bengal  2024        4

                                  type  count     amount
0                south andaman district      3      795.0
1                     nicobars district      3      565.0
2      north and middle andaman district      1      281.0
3                south andaman district     35    13651.0
4                     nicobars district      5     1448.0
...                                 ...    ...        ...
13871               alipurduar district   1023  1613143.0
13872         paschim bardhaman district   4945  7005851.0
13873                     nadia district   3807  5031294.0
13874                   birbhum district   1818  2423290.0
13875           purba medinipur district   3559  4439061.0

[13876 rows x 6 columns]
```

In [16]:
```
# LOAD THE DATA AND EXTRACTION FOR MAP USER DATAS

path=r"D:\PHONEPAY_PROJECT\first_project\pulse\data\map\user\hover\country\india\st
path = path.replace("\\", "/")
map_state_user_list=sorted(os.listdir(path))
print(map_state_user_list)

clm6 = {'state':[], 'year':[],'quarter':[],'registeredUsers':[],'appOpens':[]}

for i in map_state_user_list:
  p_i = path + i + "/"
  agg_yr = os.listdir(p_i)
  for j in agg_yr:
    p_j = p_i + j + "/"
    agg_yr_list = os.listdir(p_j)
    for k in agg_yr_list:
      p_k = p_j + k
      with open(p_k, 'r') as Data:
          D = json.load(Data)
          if "hoverData" in D["data"] and D["data"]["hoverData"]:
              for district, values in D["data"]["hoverData"].items():
                  clm6['state'].append(i)
                  clm6['year'].append(j)
                  clm6['quarter'].append(int(k.replace('.json', '')))
                  clm6['registeredUsers'].append(values.get("registeredUsers", 0))
                  clm6['appOpens'].append(values.get("appOpens", 0))


map_users=pd.DataFrame(clm6)
print(map_users)
```

```
['andaman-&-nicobar-islands', 'andhra-pradesh', 'arunachal-pradesh', 'assam', 'biha
r', 'chandigarh', 'chhattisgarh', 'dadra-&-nagar-haveli-&-daman-&-diu', 'delhi', 'go
a', 'gujarat', 'haryana', 'himachal-pradesh', 'jammu-&-kashmir', 'jharkhand', 'karna
taka', 'kerala', 'ladakh', 'lakshadweep', 'madhya-pradesh', 'maharashtra', 'manipu
r', 'meghalaya', 'mizoram', 'nagaland', 'odisha', 'puducherry', 'punjab', 'rajastha
n', 'sikkim', 'tamil-nadu', 'telangana', 'tripura', 'uttar-pradesh', 'uttarakhand',
'west-bengal']
                          state  year  quarter  registeredUsers   appOpens
0      andaman-&-nicobar-islands  2018        1              632          0
1      andaman-&-nicobar-islands  2018        1             5846          0
2      andaman-&-nicobar-islands  2018        1              262          0
3      andaman-&-nicobar-islands  2018        2              911          0
4      andaman-&-nicobar-islands  2018        2             8143          0
...                          ...   ...      ...              ...        ...
20603                west-bengal  2024        4           475688   31842355
20604                west-bengal  2024        4          1468252   80543469
20605                west-bengal  2024        4          1861738   98740305
20606                west-bengal  2024        4          1114220   73465525
20607                west-bengal  2024        4          1717074  108350777

[20608 rows x 5 columns]
```

In [17]:
```
# LOAD THE DATA AND EXTRACTION FOR TOP TRANSACTION DATAS

path=r"D:\PHONEPAY_PROJECT\first_project\pulse\data\top\transaction\country\india\s
```

```
path = path.replace("\\", "/")
top_state_trans_list=sorted(os.listdir(path))
print(top_state_trans_list)

clm7={'state':[], 'year':[],'quarter':[], 'entityName':[], 'type':[], 'count':[], '

for i in top_state_trans_list:
    p_i=path+i+"/"
    agg_yr=os.listdir(p_i)
    for j in agg_yr:
        p_j=p_i+j+"/"
        agg_yr_list=os.listdir(p_j)
        for k in agg_yr_list:
            p_k=p_j+k
            with open(p_k, 'r') as Data:
                D = json.load(Data)
            districts = D.get('data', {}).get('districts') or []
            for z in districts:
                entity = z.get('entityName')
                metric = z.get('metric', {})
                clm7['state'].append(i)
                clm7['year'].append(j)
                clm7['quarter'].append(int(k.replace('.json', '')))
                clm7['entityName'].append(entity)
                clm7['type'].append(metric.get('type'))
                clm7['count'].append(metric.get('count'))
                clm7['amount'].append(metric.get('amount'))

top_trans=pd.DataFrame(clm7)
print(top_trans)
```

```
['andaman-&-nicobar-islands', 'andhra-pradesh', 'arunachal-pradesh', 'assam', 'biha
r', 'chandigarh', 'chhattisgarh', 'dadra-&-nagar-haveli-&-daman-&-diu', 'delhi', 'go
a', 'gujarat', 'haryana', 'himachal-pradesh', 'jammu-&-kashmir', 'jharkhand', 'karna
taka', 'kerala', 'ladakh', 'lakshadweep', 'madhya-pradesh', 'maharashtra', 'manipu
r', 'meghalaya', 'mizoram', 'nagaland', 'odisha', 'puducherry', 'punjab', 'rajastha
n', 'sikkim', 'tamil-nadu', 'telangana', 'tripura', 'uttar-pradesh', 'uttarakhand',
'west-bengal']
                         state  year  quarter                entityName  \
0     andaman-&-nicobar-islands  2018        1             south andaman
1     andaman-&-nicobar-islands  2018        1                  nicobars
2     andaman-&-nicobar-islands  2018        1  north and middle andaman
3     andaman-&-nicobar-islands  2018        2             south andaman
4     andaman-&-nicobar-islands  2018        2                  nicobars
...                         ...   ...      ...                       ...
8291                west-bengal  2024        4                    howrah
8292                west-bengal  2024        4                     nadia
8293                west-bengal  2024        4                     malda
8294                west-bengal  2024        4                   hooghly
8295                west-bengal  2024        4         paschim medinipur

       type     count        amount
0     TOTAL      5688  1.256025e+07
1     TOTAL       528  1.139849e+06
2     TOTAL       442  9.316631e+05
3     TOTAL      9395  2.394824e+07
4     TOTAL      1120  3.072437e+06
...     ...       ...           ...
8291  TOTAL  67496284  1.025764e+11
8292  TOTAL  65274337  1.079320e+11
8293  TOTAL  63424613  1.024212e+11
8294  TOTAL  59676576  9.071383e+10
8295  TOTAL  58238212  8.431774e+10

[8296 rows x 7 columns]
```

In [18]:

```
# LOAD THE DATA AND EXTRACTION FOR TOP INSURANCE DATAS

path=r"D:\PHONEPAY_PROJECT\first_project\pulse\data\top\insurance\country\india\sta
path = path.replace("\\", "/")
top_state_ins_list=sorted(os.listdir(path))
print(top_state_ins_list)

clm8={'state':[], 'year':[],'quarter':[], 'entityName':[], 'type':[], 'count':[], '

for i in top_state_ins_list:
    p_i=path+i+"/"
    agg_yr=os.listdir(p_i)
    for j in agg_yr:
        p_j=p_i+j+"/"
        agg_yr_list=os.listdir(p_j)
        for k in agg_yr_list:
            p_k=p_j+k
            with open(p_k, 'r') as Data:
                D = json.load(Data)
            districts = D.get('data', {}).get('districts') or []
            for z in districts:
```

```
                entity = z.get('entityName')
                metric = z.get('metric', {})
                clm8['state'].append(i)
                clm8['year'].append(j)
                clm8['quarter'].append(int(k.replace('.json', '')))
                clm8['entityName'].append(entity)
                clm8['type'].append(metric.get('type'))
                clm8['count'].append(metric.get('count'))
                clm8['amount'].append(metric.get('amount'))

top_ins=pd.DataFrame(clm8)
print(top_ins)
```

```
['andaman-&-nicobar-islands', 'andhra-pradesh', 'arunachal-pradesh', 'assam', 'biha
r', 'chandigarh', 'chhattisgarh', 'dadra-&-nagar-haveli-&-daman-&-diu', 'delhi', 'go
a', 'gujarat', 'haryana', 'himachal-pradesh', 'jammu-&-kashmir', 'jharkhand', 'karna
taka', 'kerala', 'ladakh', 'lakshadweep', 'madhya-pradesh', 'maharashtra', 'manipu
r', 'meghalaya', 'mizoram', 'nagaland', 'odisha', 'puducherry', 'punjab', 'rajastha
n', 'sikkim', 'tamil-nadu', 'telangana', 'tripura', 'uttar-pradesh', 'uttarakhand',
'west-bengal']
                         state  year  quarter                entityName  \
0     andaman-&-nicobar-islands  2020        2                  nicobars
1     andaman-&-nicobar-islands  2020        2             south andaman
2     andaman-&-nicobar-islands  2020        3             south andaman
3     andaman-&-nicobar-islands  2020        3                  nicobars
4     andaman-&-nicobar-islands  2020        3  north and middle andaman
...                         ...   ...      ...                       ...
5603                west-bengal  2024        4         paschim bardhaman
5604                west-bengal  2024        4         paschim medinipur
5605                west-bengal  2024        4                     malda
5606                west-bengal  2024        4                     nadia
5607                west-bengal  2024        4           purba bardhaman

       type  count     amount
0     TOTAL      3      565.0
1     TOTAL      3      795.0
2     TOTAL     35    13651.0
3     TOTAL      5     1448.0
4     TOTAL      1      281.0
...     ...    ...        ...
5603  TOTAL   4945  7005851.0
5604  TOTAL   4155  5405054.0
5605  TOTAL   4042  4902098.0
5606  TOTAL   3807  5031294.0
5607  TOTAL   3595  4482879.0

[5608 rows x 7 columns]
```

In [19]:

```
# LOAD THE DATA AND EXTRACTION FOR TOP USER DATAS

path=r"D:\PHONEPAY_PROJECT\first_project\pulse\data\top\user\country\india\state\\"
path = path.replace("\\", "/")
top_state_user_list=sorted(os.listdir(path))
print(top_state_user_list)

clm9={'state':[], 'year':[],'quarter':[], 'entityName':[], 'registeredUsers':[]}
```

```
for i in top_state_ins_list:
    p_i=path+i+"/"
    agg_yr=os.listdir(p_i)
    for j in agg_yr:
        p_j=p_i+j+"/"
        agg_yr_list=os.listdir(p_j)
        for k in agg_yr_list:
            p_k=p_j+k
            with open(p_k, 'r') as Data:
                D = json.load(Data)
            districts = D.get('data', {}).get('districts') or []
            for z in districts:
                clm9['state'].append(i)
                clm9['year'].append(j)
                clm9['quarter'].append(int(k.replace('.json', '')))
                clm9['entityName'].append(z.get('name'))
                clm9['registeredUsers'].append(z.get('registeredUsers'))

top_users=pd.DataFrame(clm9)
print(top_users)
```

```
['andaman-&-nicobar-islands', 'andhra-pradesh', 'arunachal-pradesh', 'assam', 'biha
r', 'chandigarh', 'chhattisgarh', 'dadra-&-nagar-haveli-&-daman-&-diu', 'delhi', 'go
a', 'gujarat', 'haryana', 'himachal-pradesh', 'jammu-&-kashmir', 'jharkhand', 'karna
taka', 'kerala', 'ladakh', 'lakshadweep', 'madhya-pradesh', 'maharashtra', 'manipu
r', 'meghalaya', 'mizoram', 'nagaland', 'odisha', 'puducherry', 'punjab', 'rajastha
n', 'sikkim', 'tamil-nadu', 'telangana', 'tripura', 'uttar-pradesh', 'uttarakhand',
'west-bengal']
                         state  year  quarter                entityName  \
0     andaman-&-nicobar-islands  2018        1             south andaman
1     andaman-&-nicobar-islands  2018        1  north and middle andaman
2     andaman-&-nicobar-islands  2018        1                  nicobars
3     andaman-&-nicobar-islands  2018        2             south andaman
4     andaman-&-nicobar-islands  2018        2  north and middle andaman
...                         ...   ...      ...                       ...
8291                west-bengal  2024        4                   hooghly
8292                west-bengal  2024        4                     nadia
8293                west-bengal  2024        4           purba medinipur
8294                west-bengal  2024        4         paschim medinipur
8295                west-bengal  2024        4         paschim bardhaman

      registeredUsers
0                5846
1                 632
2                 262
3                8143
4                 911
...               ...
8291          1957356
8292          1861738
8293          1717074
8294          1522845
8295          1468252

[8296 rows x 5 columns]
```

In [20]:
```
# IMPORT LIBRARIES FOR CONNECTING THE SQL

import mysql.connector
import pandas as pd
import os
from sqlalchemy import create_engine, text, MetaData, Table, Column, Integer, Strin
```

In [21]:
```
# SQL INSERTION

username = "root"
password = "your_password"
host = "localhost"
database = "phonepe_db"

engine = create_engine(f"mysql+pymysql://{username}:{password}@{host}/{database}")
```

In [22]:
```
# CLM1 TABLE INSERTION TO SQL

agg_trans.to_sql(
    name="aggregated_transactions",
    con=engine,
    if_exists="replace",   # use 'append' if you don't want to delete existing tabl
    index=False,
    chunksize=1000,
    method="multi"
)

print("✅ DataFrame successfully written to MySQL!")
```

```
✅ DataFrame successfully written to MySQL!
```

In [23]:
```
# CLM2 TABLE INSERTION TO SQL

agg_ins.to_sql(
    name="aggregated_insurance",
    con=engine,
    if_exists="replace",   # use 'append' if you don't want to delete existing tabl
    index=False,
    chunksize=1000,
    method="multi"
)

print("✅ DataFrame successfully written to MySQL!")
```

```
✅ DataFrame successfully written to MySQL!
```

In [24]:
```
# CLM3 TABLE INSERTION TO SQL

agg_users.to_sql(
    name="aggregated_users",
    con=engine,
    if_exists="replace",   # use 'append' if you don't want to delete existing tabl
    index=False,
    chunksize=1000,
    method="multi"
)
```

```
print("✅ DataFrame successfully written to MySQL!")
```

```
✅ DataFrame successfully written to MySQL!
```

In [25]:
```
# CLM4 TABLE INSERTION TO SQL

map_trans.to_sql(
    name="map_transactions",
    con=engine,
    if_exists="replace",   # use 'append' if you don't want to delete existing tabl
    index=False,
    chunksize=1000,
    method="multi"
)

print("✅ DataFrame successfully written to MySQL!")
```

```
✅ DataFrame successfully written to MySQL!
```

In [26]:
```
# CLM5 TABLE INSERTION TO SQL

map_ins.to_sql(
    name="map_insurance",
    con=engine,
    if_exists="replace",   # use 'append' if you don't want to delete existing tabl
    index=False,
    chunksize=1000,
    method="multi"
)

print("✅ DataFrame successfully written to MySQL!")
```

```
✅ DataFrame successfully written to MySQL!
```

In [27]:
```
# CLM6 TABLE INSERTION TO SQL

map_users.to_sql(
    name="map_users",
    con=engine,
    if_exists="replace",   # use 'append' if you don't want to delete existing tabl
    index=False,
    chunksize=1000,
    method="multi"
)

print("✅ DataFrame successfully written to MySQL!")
```

```
✅ DataFrame successfully written to MySQL!
```

In [28]:
```
# CLM7 TABLE INSERTION TO SQL

top_trans.to_sql(
    name="top_transactions",
    con=engine,
    if_exists="replace",   # use 'append' if you don't want to delete existing tabl
    index=False,
    chunksize=1000,
```

```
    method="multi"
)

print("✅ DataFrame successfully written to MySQL!")
```

```
✅ DataFrame successfully written to MySQL!
```

In [29]:

```
# CLM8 TABLE INSERTION TO SQL

top_ins.to_sql(
    name="top_insurance",
    con=engine,
    if_exists="replace",   # use 'append' if you don't want to delete existing tabl
    index=False,
    chunksize=1000,
    method="multi"
)

print("✅ DataFrame successfully written to MySQL!")
```

```
✅ DataFrame successfully written to MySQL!
```

In [30]:

```
# CLM9 TABLE INSERTION TO SQL

top_users.to_sql(
    name="top_users",
    con=engine,
    if_exists="replace",   # use 'append' if you don't want to delete existing tabl
    index=False,
    chunksize=1000,
    method="multi"
)

print("✅ DataFrame successfully written to MySQL!")
```

```
✅ DataFrame successfully written to MySQL!
```

In [ ]: